CliniScan: AI-Based Chest X-Ray Abnormality Classification

Vibha Venu

1. Project Description

CliniScan is a deep learning-based system developed to automatically detect abnormalities in chest X-ray images. Chest X-rays are one of the most commonly used imaging techniques in medical diagnosis, especially for detecting lung-related diseases. However, manually examining large numbers of X-ray images can be time-consuming and requires significant expertise.

The main objective of this project is to build a deep learning model that can analyse chest X-ray images and predict the presence of different abnormalities. The system uses convolutional neural networks (CNNs) to learn patterns from medical images and classify them into multiple disease categories.

Unlike traditional classification systems that assign only one label to an image, this project uses **multi-label classification**, where a single X-ray image may contain more than one abnormality. The model therefore predicts the probability of each disease independently.

The project also aims to build a complete pipeline that includes dataset preparation, preprocessing, model training, evaluation, and visualization of training results.

2. Dataset Used

The dataset used in this project consists of chest X-ray images along with corresponding annotation files that indicate the presence of abnormalities in the images.

Each image in the dataset has an associated **text file (.txt)** that contains annotation information. These label files store the details of abnormalities detected in the image.

To organize the data efficiently for model training and evaluation, the dataset is structured into separate folders for training, validation, and testing.

Each image file has a corresponding label file with the same name but a **.txt extension**.

Example:

images/train/image001.png
labels/train/image001.txt

Each .txt file contains annotation information for the objects present in the image. Every line in the file represents one abnormality and includes the class identifier along with positional information of the abnormality within the image.

Example annotation format:

class_id x_center y_center width height

These values represent the location and size of the abnormality region in the image. The coordinates are normalized relative to the image dimensions.

This dataset structure allows the model to learn patterns associated with different chest abnormalities and supports both classification and detection tasks within the CliniScan system.

3. Environment Setup

The project was implemented using **Python** with the **PyTorch deep learning framework**. The development environment used was **Visual Studio Code**.

The following libraries were used during development:

- **PyTorch (torch, torchvision)** – for building and training deep learning models
- **NumPy** – for numerical computations
- **Pandas** – for handling structured data
- **Matplotlib** – for visualizing training results
- **Scikit-learn** – for computing evaluation metrics such as AUC-ROC and F1-score
- **Pillow (PIL)** – for loading and processing image files

A virtual environment was created to manage dependencies and maintain consistency across the project.

The training process can run on either **CPU or GPU**, depending on hardware availability. I have used CPU.

4. Data Exploration

Before training the model, the dataset was explored to ensure that the images and annotations were correctly structured.

The dataset folders were inspected to confirm that each image had a corresponding annotation file. This helped verify that the dataset was complete and correctly organised.

Sample images were loaded and displayed to check image quality and resolution. This ensured that the images were readable and suitable for training.

The annotation files were also inspected to confirm that the bounding box values and class identifiers were correctly formatted.

A separate data verification script was used to detect potential issues such as:

- Missing label files
- Empty annotation files
- Image size mismatches

These checks help prevent errors during model training and ensure that the dataset is correctly prepared.

5. Data Preprocessing

Raw medical images cannot be directly used for training deep learning models. Therefore, several preprocessing steps were applied to prepare the dataset.

Image Loading

Images were loaded using the Pillow library and converted into RGB format. This ensures consistent color channels across all images.

Image Resizing

All images were resized to a fixed resolution of **256 × 256 pixels**. Using a fixed image size allows the model to process images efficiently during training.

Data Augmentation

To improve the model's ability to generalize to unseen data, several augmentation techniques were applied during training:

- Random horizontal flipping
- Random rotation
- Brightness and contrast adjustments

These transformations help simulate variations that may occur in real-world medical images.

Normalization

Images were normalized using standard ImageNet statistics. Normalization helps stabilize the training process and improves model performance when using pretrained networks.

Tensor Conversion

The processed images and labels were converted into PyTorch tensors before being passed into the neural network.

6. Proposed System

The CliniScan system follows a deep learning pipeline for analysing chest X-ray images.

The process begins with a raw chest X-ray image that is provided as input to the system. The image first undergoes preprocessing, where it is resized, normalized, and optionally augmented.

The processed image is then passed into a convolutional neural network for feature extraction.

The system supports multiple pretrained architectures, including:

- **ResNet50**
- **EfficientNet-B0**
- **EfficientNet-B2**
- **DenseNet121**

These networks extract complex visual features from the input image and pass them through a classification layer that predicts the presence of multiple abnormalities.

Since chest X-rays may contain more than one abnormality, the system uses **multi-label classification**, where each class is predicted independently.

Detection Module (In Progress)

While the current implementation focuses on predicting the presence of abnormalities, the next stage of the project aims to extend the system with an abnormality detection module.

This module will use the annotation information to identify and highlight the exact locations of abnormalities in the X-ray image using bounding boxes.

Object detection architectures such as **YOLO** or **Faster R-CNN** are being explored for this purpose.

At the time of writing this report, the **detection model is currently under development** and will be integrated into the system in future iterations.

System Architecture:

7. Training Pipeline

The dataset was divided into three subsets:

- **Training set – 70%**
- **Validation set – 15%**
- **Test set – 15%**

The training set is used to train the neural network, while the validation set is used to monitor performance during training. The test set is used for final evaluation.

The model was trained using the **Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss)** function, which is suitable for multi-label classification tasks.

The **Adam optimizer** was used to update model parameters during training with a learning rate of **0.001**.

A learning rate scheduler was also used to automatically reduce the learning rate when the validation loss stopped improving.

Model performance was evaluated using the following metrics:

- **AUC-ROC (Area Under the Receiver Operating Characteristic Curve)**
- **F1-score**

During training, the best-performing model was saved automatically based on validation performance.

Training progress was also visualized using graphs showing training loss, validation loss, and evaluation metrics across epochs.

8. Conclusion

This project demonstrates how deep learning techniques can be applied to medical imaging for automated chest X-ray analysis.

The CliniScan system successfully implements a complete pipeline including dataset preparation, preprocessing, model training, and evaluation. By using pretrained convolutional neural networks, the system is able to learn meaningful patterns from chest X-ray images and predict multiple abnormalities.

Although the current version of the project focuses on multi-label classification, the next stage will integrate an abnormality detection module capable of localizing disease regions within the X-ray image.

Such systems have the potential to support radiologists by assisting in faster and more accurate diagnosis of medical conditions.